

EXP 19:

Determination and plotting of Antenna Efficiency.

Aim: To determine and plot Antenna Efficiency and to Estimate the efficiency based on radiated and input power.

INTPUT Code:

```
"Radiated Power:"
0.2500000
"Antenna Efficiency (%):"
82.5
-->
```

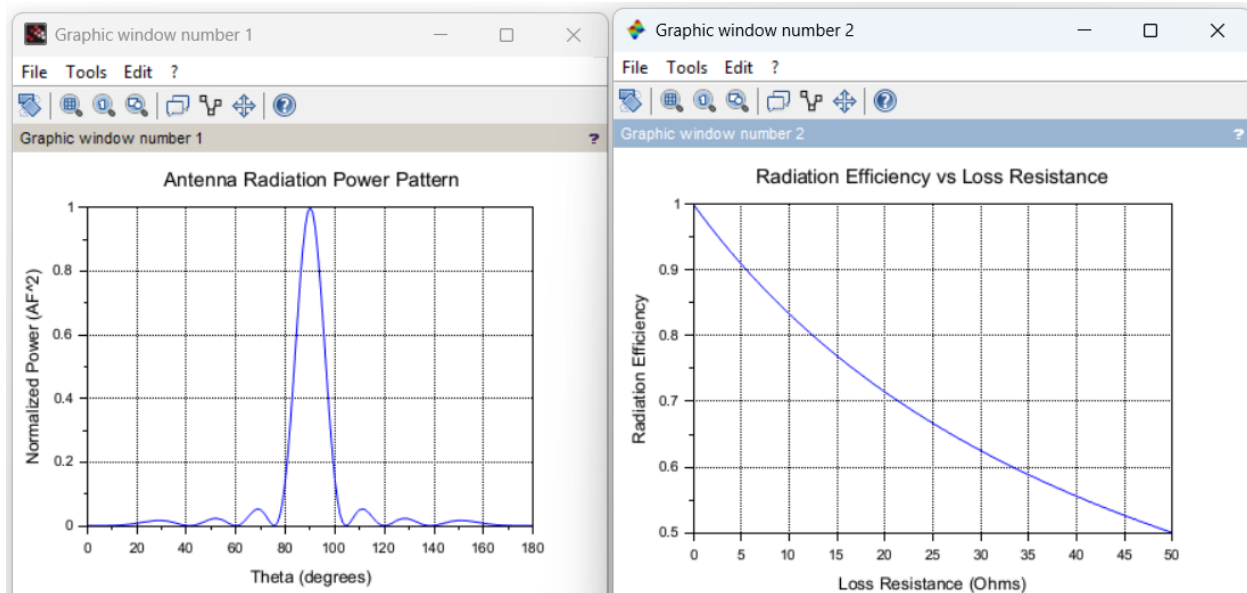
```
*exp333.scd X
1 clc;
2 clear;
3 close;
4 theta_deg = 0:1:180;
5 theta = theta_deg * %pi / 180;
6 N = 8;
7 d = 0.5;
8 beta = 0;
9 psi = 2 * %pi * d * cos(theta) + beta;
10 AF = zeros(theta);
11 for i = 1:length(theta)
12     if abs(sin(psi(i)/2)) < 0.000001 then
13         AF(i) = N;
14     else
15         AF(i) = abs(sin(N*psi(i)/2) / sin(psi(i)/2));
16     end
17 end
18 AF = AF / max(AF);
19 Power = AF.^2;
20 dtheta = theta(2) - theta(1);
21 radiated_power = sum(Power .* sin(theta)) * dtheta;
22 input_power = 1;
23 R_r = 50;
24 R_loss = 10;
25 Gamma_selected = 0.1;
26 radiation_efficiency = R_r / (R_r + R_loss);
27 coupling_efficiency = 1 - Gamma_selected^2;
28 total_efficiency = radiation_efficiency * coupling_efficiency;
29 efficiency_percent = total_efficiency * 100;
30 disp("Radiated Power:");
31 disp(radiated_power);
32 disp("Antenna Efficiency (%):");
33 disp(efficiency_percent);
34 scf(1);
35 plot(theta_deg, Power);
36 xlabel("Theta (degrees)");
37 ylabel("Normalized Power (AF^2)");
38 title("Antenna Radiation Power Pattern");
39 xgrid();
40 R_l = linspace(0, 50, 500);
41 e_r = R_r ./ (R_r + R_l);
```

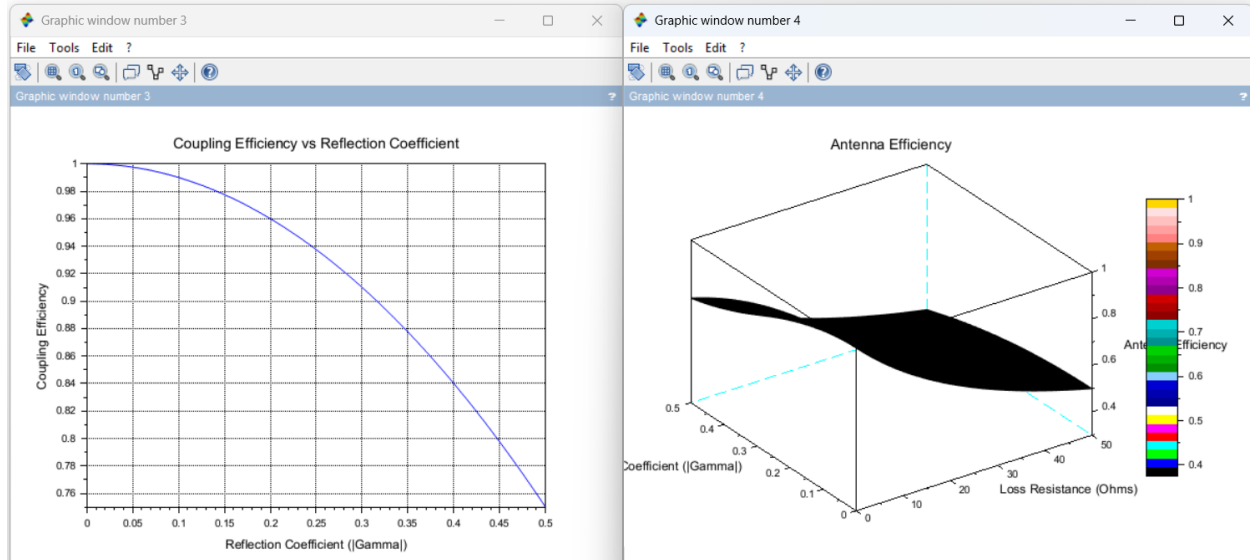
```

42 scf(2);
43 plot(R_l, e_r);
44 xlabel("Loss-Resistance-(Ohms)");
45 ylabel("Radiation-Efficiency");
46 title("Radiation-Efficiency-vs-Loss-Resistance");
47 xgrid();
48 Gamma = linspace(0, 0.5, 500);
49 e_c = 1 - Gamma.^2;
50 scf(3);
51 plot(Gamma, e_c);
52 xlabel("Reflection-Coefficient-(|Gamma|)");
53 ylabel("Coupling-Efficiency");
54 title("Coupling-Efficiency-vs-Reflection-Coefficient");
55 xgrid();
56 [RL, GAMMA] = meshgrid(R_l, Gamma);
57 ER = R_r ./ (R_r + RL);
58 EC = 1 - GAMMA.^2;
59 EA = ER .* EC;
60 scf(4);
61 surf(R_l, Gamma, EA);
62 xlabel("Loss-Resistance-(Ohms)");
63 ylabel("Reflection-Coefficient-(|Gamma|)");
64 zlabel("Antenna-Efficiency");
65 title("Antenna-Efficiency");
66 colorbar();

```

OUTPUT:





CASE STUDY 1: Antenna Efficiency in Satellite Communication Systems

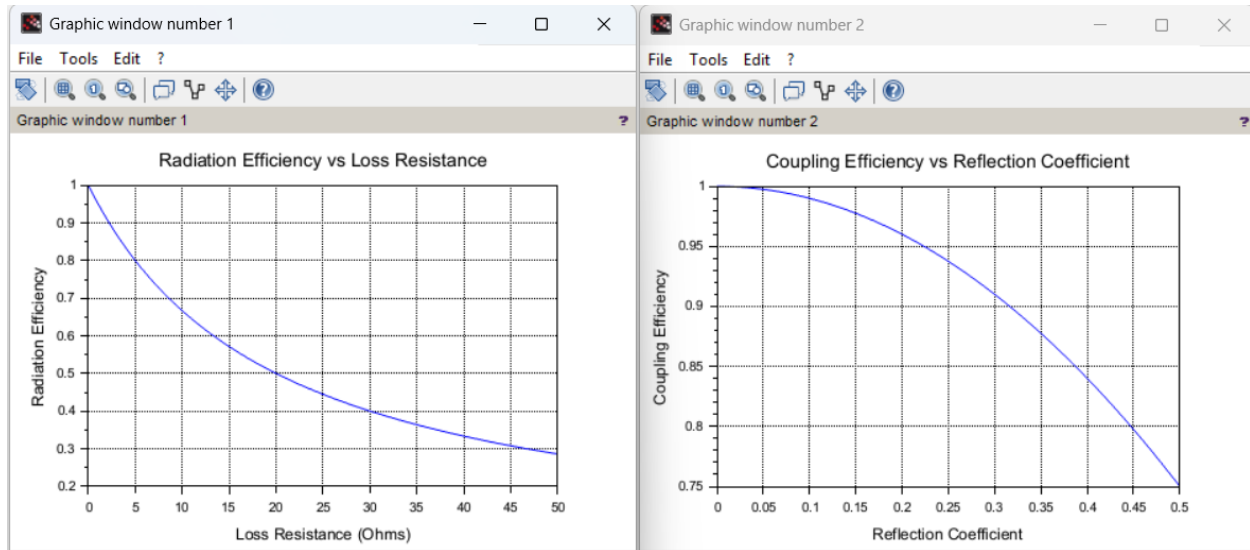
```

"Radiation Efficiency (%)" =
66.666667
"Coupling Efficiency (%)" =
99.
"Overall Antenna Efficiency (%)" =
66.000000
-->

file edit format options window execute
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
*exp333.sci
1 clc;
2 clear;
3 close;
4 frequency = 12e9;
5 diameter = 2.5;
6 Rl = 10;
7 Rr = 20;
8 Z0 = 50;
9 Gamma = 0.1;
10 er = Rr ./ (Rr + Rl);
11 ec = 1 - Gamma^2;
12 ea = er * ec;
13 er_percent = er * 100;
14 ec_percent = ec * 100;
15 ea_percent = ea * 100;
16 disp("Radiation Efficiency (%) =");
17 disp(er_percent);
18 disp("Coupling Efficiency (%) =");
19 disp(ec_percent);
20 disp("Overall Antenna Efficiency (%) =");
21 disp(ea_percent);
22 Rloss = linspace(0, 50, 100);
23 er_plot = Rr ./ (Rr + Rloss);
24 scf(1);
25 plot(Rloss, er_plot);
26 xlabel("Loss Resistance (Ohms)");
27 ylabel("Radiation Efficiency");
28 title("Radiation Efficiency vs Loss Resistance");
29 xgrid();
30 Gamma_plot = linspace(0, 0.5, 100);
31 ec_plot = 1 - Gamma_plot.^2;
32 scf(2);
33 plot(Gamma_plot, ec_plot);
34 xlabel("Reflection Coefficient");
35 ylabel("Coupling Efficiency");
36 title("Coupling Efficiency vs Reflection Coefficient");
37 xgrid();

```

OUTPUT:



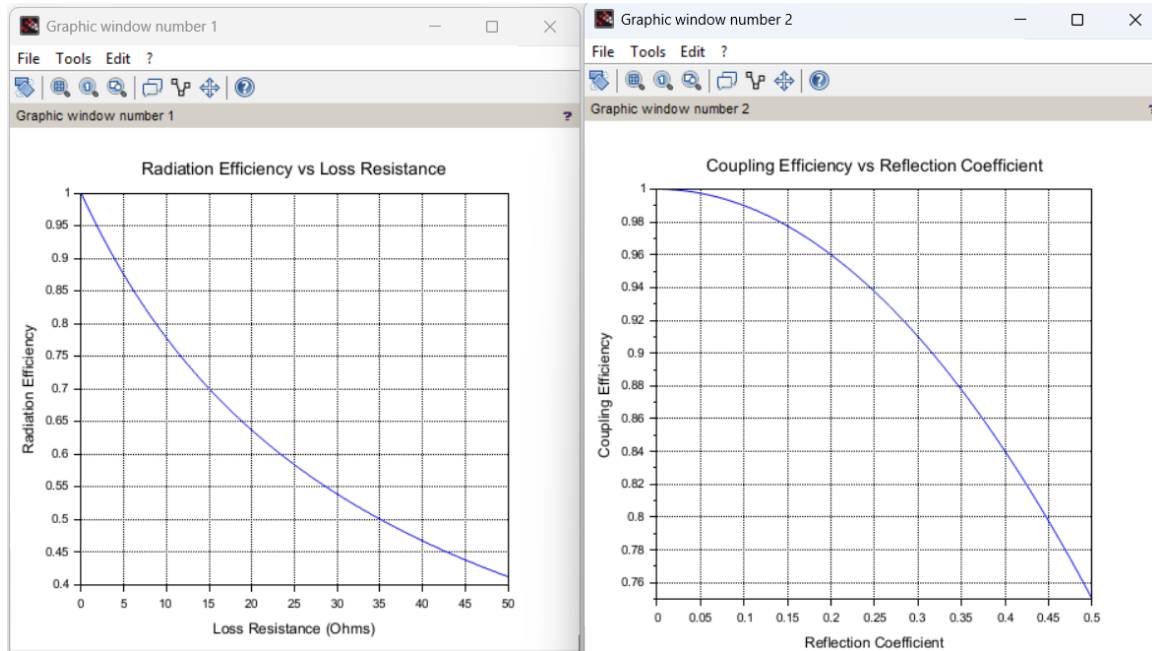
Case Study 2: Antenna Efficiency in Wireless LAN

```
"Radiation Efficiency (%) ="
87.5
"Coupling Efficiency (%) ="
99.960000
"Overall Antenna Efficiency (%) ="
87.465

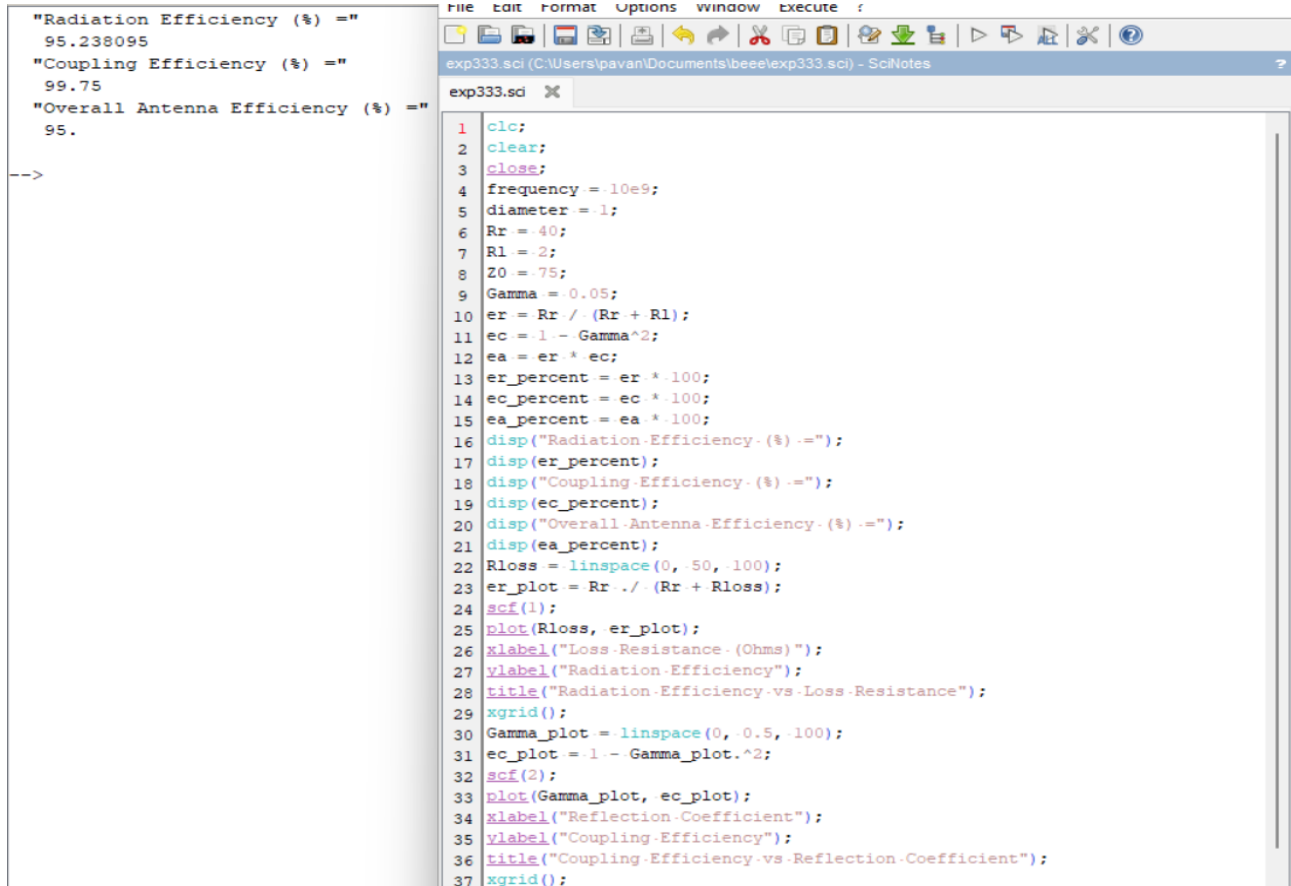
-->

file Edit Format Options Window Execute :
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
exp333.sci X
1 clc;
2 clear;
3 close;
4 frequency = 2.4e9;
5 gain_dBi = 5;
6 Z0 = 50;
7 Rl = 5;
8 Rr = 35;
9 Gamma = 0.02;
10 er = Rr ./ (Rr + Rl);
11 ec = 1 - Gamma^2;
12 ea = er * ec;
13 er_percent = er * 100;
14 ec_percent = ec * 100;
15 ea_percent = ea * 100;
16 disp("Radiation Efficiency (%) .=");
17 disp(er_percent);
18 disp("Coupling Efficiency (%) .=");
19 disp(ec_percent);
20 disp("Overall Antenna Efficiency (%) .=");
21 disp(ea_percent);
22 Rloss = linspace(0, 50, 100);
23 er_plot = Rr ./ (Rr + Rloss);
24 scf(1);
25 plot(Rloss, er_plot);
26 xlabel("Loss Resistance (Ohms)");
27 ylabel("Radiation Efficiency");
28 title("Radiation Efficiency vs Loss Resistance");
29 xgrid();
30 Gamma_plot = linspace(0, 0.5, 100);
31 ec_plot = 1 - Gamma_plot.^2;
32 scf(2);
33 plot(Gamma_plot, ec_plot);
34 xlabel("Reflection Coefficient");
35 ylabel("Coupling Efficiency");
36 title("Coupling Efficiency vs Reflection Coefficient");
37 xgrid();
```

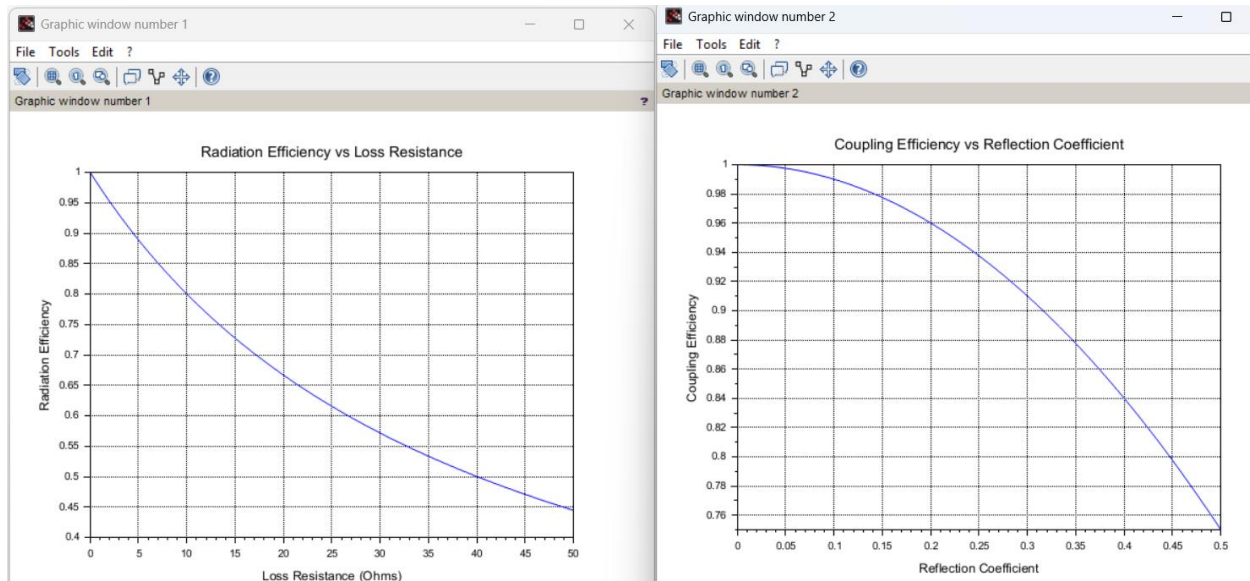
OUTPUT:



Case Study 3: Antenna Efficiency in Radar Systems



OUTPUT:



Case Study 4: Efficiency of a Microstrip Patch Antenna for Mobile Communication

```
"Radiation Efficiency (%) ="
95.238095
"Coupling Efficiency (%) ="
99.75
"Overall Antenna Efficiency (%) ="
95.

-->
```

```
File Edit Format Options Window Execute :
exp333.sci (C:\Users\pavan\Documents\bsee\exp333.sci) - SciNotes
*exp333.sci X
1 clc;
2 clear;
3 close;
4 frequency = 1.8e9;
5 patch_length = 5;
6 patch_width = 5;
7 Rr = 0.15;
8 Rl = 0.5;
9 Z0 = 50;
10 Gamma = 0.15;
11 er = Rr / (Rr + Rl);
12 ec = 1 - Gamma^2;
13 ea = er * ec;
14 er_percent = er * 100;
15 ec_percent = ec * 100;
16 ea_percent = ea * 100;
17 disp("Radiation Efficiency (%) =");
18 disp(er_percent);
19 disp("Coupling Efficiency (%) =");
20 disp(ec_percent);
21 disp("Overall Antenna Efficiency (%) =");
22 disp(ea_percent);
23 Rloss = linspace(0, 50, 100);
24 er_plot = Rr ./ (Rr + Rloss);
25 scf(1);
26 plot(Rloss, er_plot);
27 xlabel("Loss Resistance (Ohms)");
28 ylabel("Radiation Efficiency");
29 title("Radiation Efficiency vs Loss Resistance");
30 xgrid();
31 Gamma_plot = linspace(0, 0.5, 100);
32 ec_plot = 1 - Gamma_plot.^2;
33 scf(2);
34 plot(Gamma_plot, ec_plot);
35 xlabel("Reflection Coefficient");
36 ylabel("Coupling Efficiency");
37 title("Coupling Efficiency vs Reflection Coefficient");
38 xgrid();
```


OUTPUT:

